

Srikar Amirneni

805-550-0418 | samirnen@calpoly.edu | linkedin.com/in/srikar-amirneni | srikaramirneni.vercel.app

Education

California Polytechnic State University-San Luis Obispo

June 2026

B.S. in Computer Science

San Luis Obispo, CA

- Coursework: Data Structures & Algorithms, Operating Systems (C), Databases (SQL), Programming Languages (Racket), OOP (Java)

Experience

AI Engineering Intern

March 2025 – September 2025

SmallChess

San Luis Obispo, CA

- Modernized a legacy chess app with **React Native (Expo)**, reducing **mobile load times by 20%**
- Developed backend APIs for **500+ annotated games**, improving reliability and cutting **fetch latency by 35%**
- Engineered an **AI-assisted training** feature integrating **Pinecone** vector DB, **LangChain** embeddings, and a custom **MCTS** engine to generate contextual move explanations
- Assembled an Agentic AI pipeline** that turns user prompts into chess-variant **rule sets** with automatic move validation

Software Engineering Consultant

April 2025 – August 2025

Harrow Services

San Luis Obispo, CA

- Designed and launched a client website and app, boosting **SEO** rankings and enabling **live player tracking**
- Delivered scalable infrastructure and technical strategy to support **new digital revenue streams**
- Collaborated with cross-functional teams to implement **payment integration** and **automated scheduling**
- Optimized **AWS** deployment, reducing **infrastructure costs by 15%** while maintaining high reliability

React Developer

January 2025 – May 2025

MindSprout

San Luis Obispo, CA

- Launched **React**-based UI components and 4 new mini-games, increasing engagement in a gambling rehabilitation app
- Partnered with psychologists to align features with therapeutic goals, contributing to **user retention** and relapse prevention
- Implemented **secure authentication** using **JWT** and **bcrypt** and **progress tracking**, supporting **hundreds of active users**
- Streamlined **Agile sprints** and **code reviews**, enabling on-time feature releases and higher code quality

Unity Developer

October 2023 – May 2025

California Cybersecurity Institute

San Luis Obispo, CA

- Built a global **Unity3D + C#** Capture the Flag platform for **500+ participants**, boosting **FPS by 3.5×** and improving playability
- Resolved **40+** live support requests and mentored teams, ensuring smooth execution of competitions and bug fixes on time
- Created **CI/CD** pipelines for **AWS S3** deployments, reducing deployment time significantly and improved version control
- Optimized **AppStream** performance with AWS engineers, enabling access on low-power devices like Chromebooks

Research Experience

Research Assistant

October 2025 – Present

California Polytechnic State University, San Luis Obispo

San Luis Obispo, CA

- Set up a reproducible **Debian Linux VM** lab for safe process-injection experiments with scripted runs and logs
- Prototyped and evaluated **ptrace**, **/proc** overwrite, and **LD_PRELOAD** paths, validating **overwrite → execute → recover**
- Reviewing literature and collaborating with **JPL** and **SRI** to prioritize novel vectors and assess **Linux** defenses

Technical Projects

Ice Breaker Python, Flask, LangChain, Tavily Search

June 2025

- Programmed a **Flask** app that uses **LangChain** with **Tavily Search** to locate a target's LinkedIn profile and summarize public information
- Generated outreach insights ("icebreakers") from role, recent posts, and shared topics to guide first-contact messages
- Added **input validation**, **rate limiting**, and structured logging with clear error handling for reliable searches

SpreadIt React, Next.js, PostgreSQL

January 2025 – March 2025

- Led a **4-person Agile** build of a gamified sandwich-rating app; responsible for feature coding, testing, code reviews, and documentation
- Reduced page load time by **20%** through profiling, caching, and API optimization; improved Core Web Vitals
- Scaled to **1,000+ daily interactions** with **PostgreSQL** and **RBAC**; integrated auth and added **Jest** test coverage

Skills

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, C#, Racket, HTML/CSS

Frameworks/Libraries: React, React Native (Expo), Next.js, Node.js, Flask, LangChain, NumPy, pandas, Jest

Systems/Tools: PostgreSQL, REST APIs, Docker, Linux/Unix, GitHub Actions, AWS, Git, LLVM, VS Code, Unity